

GUIDED LESSON · LEARN BY EXPERIMENTING

Explain Your Observation & Check Your Understanding

Two wrap-up topics for total beginners: turn what you saw in the $H \rightarrow Z \rightarrow H$ circuit into a solid explanation, then test that understanding against five quick questions.

07

Explain Your Observation

Turning what you saw into a solid reason why

Back in Section 05 you ran this circuit and got a guaranteed $|1\rangle$, never $|0\rangle$. The lesson's prompt is simple to state and easy to answer badly: “Why did that happen?” Here's a quick recap of the circuit, then how to build a complete answer.

q0



From Section 05 — the circuit you're explaining.

What a complete answer needs to mention

- **What Z did on its own** — flipped the $|1\rangle$ amplitude's sign, but changed no probability by itself (Section 02).
- **That the second H doesn't just “undo” the first** — it recombines both branches into each possible output, the way any H does (Section 04).
- **Why the branches disagree for $|0\rangle$ but agree for $|1\rangle$** — the sign Z introduced makes the two $|0\rangle$ contributions cancel and the two $|1\rangle$ contributions add (Section 06).
- **The conclusion** — the final probabilities come from the combined amplitude after all three gates, not from reasoning about each gate's probability in isolation.

MODEL ANSWER

“Z doesn't change any probability by itself — it only flips the sign of the $|1\rangle$ amplitude. But that sign matters once the second H arrives, because H recombines the $|0\rangle$ and $|1\rangle$ branches into both outputs. For the $|0\rangle$ output, the two branches now carry opposite signs and cancel exactly — destructive interference. For the $|1\rangle$ output, they carry the same sign and reinforce exactly — constructive interference. So the circuit doesn't just pass probabilities through gate by gate; it's the final interference between amplitudes that decides the answer, and that's why the outcome is a guaranteed 1 instead of another 50/50 split.”

A COMMON MISTAKE TO AVOID

“Z doesn’t change probability, so nothing about the outcome should change” sounds reasonable but stops too early — it only accounts for Z, not the second H that comes after it. Always follow the state all the way to the *last* gate before reasoning about probabilities.

TAKEAWAY

A good explanation names the invisible step (Z’s phase flip), the recombining step (the second H), and the visible result (interference) — in that order.

08**Check Your Understanding**

Five questions covering everything so far

Work through these without looking back if you can — they cover amplitude, phase, the Z gate, the H gate, and interference together.

Q1. If a qubit’s amplitude for $|1\rangle$ is -0.6 , what’s the probability of measuring 1?

ANSWER

Square the magnitude: $(-0.6)^2 = \mathbf{0.36}$, a 36% chance. The minus sign disappears once you square it — which is exactly why phase can’t be read off probability alone.

Q2. True or false: multiplying an entire statevector by -1 changes what you’ll measure.

ANSWER

False. Multiplying every amplitude by the same factor is a global phase — it never affects any measurement, now or after further gates.

Q3. What does the Z gate do to the state $(|0\rangle + |1\rangle)/\sqrt{2}$?

ANSWER

It flips the sign of only the $|1\rangle$ amplitude, giving $(|0\rangle - |1\rangle)/\sqrt{2}$ — same probabilities (0.5, 0.5), different relative phase.

Q4. In the $H \rightarrow Z \rightarrow H$ circuit, why is the outcome a guaranteed 1 instead of another 50/50 split?

ANSWER

The final H recombines two path contributions for each outcome. Because Z flipped a sign, the two contributions to $|0\rangle$ arrive with opposite sign and cancel (destructive interference), while the two contributions to $|1\rangle$ arrive with the same sign and add (constructive interference).

Q5. What single gate is $H \cdot Z \cdot H$ equivalent to?

ANSWER

The **X gate** — the quantum bit-flip (NOT) gate.

TAKEAWAY

Across every question here, one idea does all the work: relative phase is invisible until two amplitudes recombine — and then it completely controls whether they add or cancel.